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Outcome Analysis of Visual Acuity and Side Effect after Ruthenium-106 Plaque Brachytherapy for Medium-sized Choroidal Melanoma

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Abstract:

PURPOSE: The purpose of this study is to report on treatment outcomes for medium-sized choroidal melanoma treated with Ruthenium-106 (Ru-106) plaque brachytherapy.

METHODS: A retrospective case series of 28 patients received Ru-106 brachytherapy treatment for choroidal melanoma. The prescribed tumor dose was 85 Gy to a depth of 5 mm.

RESULTS: Median follow-up was 31.2 months. At 12 and 24-month postirradiation, the best corrected visual acuity $\geq 20/70$ (LogMar ≥ -0.54) was 53.8% and 64.2%, respectively. Median time to tumor regression was estimated to be 10 months (95% CI = 9–18 months), with 100% of response rate by 32 months. Radiation-induced side effects were limited, and there were no postradiation enucleations.

CONCLUSIONS: The majority of patients maintained good visual acuity, with no enucleations and minimal side effects. In this cohort, the Ru-106 plaque brachytherapy proved to be an efficacious and safe treatment option for patients with medium-sized choroidal melanomas with a maximal tumor height of 5 mm.

Keywords:

Choroidal melanoma, plaque brachytherapy, Ruthenium-106

Introduction

At present, for patients diagnosed with choroidal melanoma, there are several treatment modalities available, depending on the extent, and the location of tumor at diagnosis. Such modalities include photocoagulation, transpupillary thermotherapy, plaque radiotherapy, charged-particle irradiation, stereotactic photon radiotherapy, local resection, enucleation, chemotherapy, and immunotherapy.^[1-5] In addition, some investigators suggest a watch and wait period depending on tumor size and tumors that were deemed stable.^[6,7] More

recent reports indicate that early treatment is associated with prevention of metastasis and early death.^[8] For a long-time enucleation was the standard treatment for all choroidal melanomas. That has changed with the advances of radiotherapy and other conservative therapies, with these treatment modalities achieving survival rates comparable to those of enucleation for small- and medium-sized melanomas.^[9]

Plaque radiotherapy has emerged as a treatment option for those patients diagnosed with small- to medium-sized choroidal melanomas. A large trial, the Collaborative Ocular Melanoma Study (COMS), examined the efficacy of treating medium-sized choroidal melanoma (unilateral tumor

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with an apical height of 2.5–10.0 mm and a maximum basal tumor diameter of 16.0 mm or less) with Iodine-125 (I-125) brachytherapy versus enucleation.^[10] This long-term study demonstrated that there were no survival differences observed between patients treated with I-125 brachytherapy versus those treated with enucleation.^[11] Although I-125 brachytherapy has become the primary choice for plaque brachytherapy of choroidal melanomas in many institutes, other isotopes, such as Ruthenium-106 (Ru-106), have also been investigated for the treatment of medium-sized tumors.^[12-14] As a β -emitting isotope, Ru-106 has a limited depth of penetration, which is an advantage for choroidal melanoma treatment to spare surrounding normal tissue and an ideal radioisotope for use in medium-sized melanomas of ≤ 5 mm in height. In this paper, we present the outcomes of a single-center cohort of consecutive patients with medium-sized choroidal melanoma treated with Ru-106 plaque brachytherapy.

Methods

Patients

This retrospective study included 28 consecutive patients treated with Ru-106 plaque brachytherapy at the Department of Radiation Oncology, Montreal General Hospital of McGill University Health Centre, for choroidal melanoma between 2001 and 2010. The review was done after obtaining a written consent from the local institution according to internal ethics guidelines for retrospective reviews.

Tumors

All tumors were medium-sized choroidal and ciliary body melanomas with a maximum apical height of 5.25 mm and 16 mm in the longest basal diameter for Ru-106 plaque dosing penetration, as described previously.^[11] The diagnosis was determined based on clinical criteria, including: tumor location, maximum basal tumor diameter, and tumor thickness (determined by A and B scan ultrasonography). All patients had medical history and a complete physical examination done at the first consultation visit. Before initiating treatment, patients had a metastatic workup that included blood tests, chest radiograph, and abdominal ultrasound.

Treatment

All tumors were treated with a Ru-106 plaque, with a plaque diameter varying from 12 to 20 mm. The plaque was inserted by the ophthalmologic surgeons posterior to the tumor and position confirmed with intraoperative ultrasound. The prescription was according to the revised COMS guidelines, and the calibration certificates for all plaques were traceable to the NIST national standards laboratory through BEBIG certificate 001706. The prescribed tumor dose was 85 Gy to a depth of

5 mm. Radiation dosage to the tumor apex, sclera, and prescription point were determined by both a treatment planning system, "Plaque Simulator Version 5.0.7," and a hand check using the calibration certificate.

Follow-up and outcome assessment

Patients were seen every 4 months after irradiation for the 1st year and then every 6 months thereafter by both ophthalmology and radiation oncology. The follow-up visits involved assessment of best-corrected visual acuity through Snellen chart, slit-lamp examination, applanation tonometry, and A and B ultrasonography. Visual acuity worse than 20/200 (LogMar-1.00) was recorded either as counting fingers, hand movements, or light perception.

Statistical methods

Summary statistics were used to describe patients, treatments, tumors, and outcomes. The Kaplan–Meier method was used to estimate the proportion of patients who were regression-free from the date of Ru-106 plaque insertion. Those patients who did not have tumor regression were censored at the date of the last follow-up. Cox proportional hazards regression was used to investigate for prognostic factors of time to regression. Logistic regression was used to investigate for factors prognostic of visual acuity at 1 year. Variables considered in the analysis included: age, gender, history of diabetes mellitus or hypertension, use of notched plaque, size of lesion, retinal invasion, lesion location, total radiation dose to the prescription point, radiation dose to apex, and treatment length. Analyses were performed using SAS v9.1 (SAS Institute, Cary, NC, USA) and all tests were two-sided with a $P \leq 0.05$ was considered statistically significant.

Results

Patient population and tumors

Table 1 shows a summary of patient and tumor characteristics. Tumor apical height ranged from 2.3 to 5.25 mm, with a median height of 3.5 mm. Two patients had a tumor height >5 mm; one with height 5.20 mm and a second at 5.25 mm. Twenty eyes had a dome-shaped tumor; three were flat tumors at diagnosis; and one patient had a collar button shape. Of patients with documented tumor size postirradiation (total = 24), all the tumors had response to the treatment.

Two patients had no postradiation treatment follow-up. Of the remaining 26 patients, follow-up ranged from 1.5 to 91.5 months, with a median of 31.2 months. Median time to tumor regression was estimated to be 10 months (95% CI = 9–18 months), and is shown in Figure 1. Although no factor was observed to be statistically significant, Table 2 describes the potential

prognostic value of selected factors on time to regression.

Dosage

The treatment time ranged from 57.1 to 170.9 h, with a median of 98.2 h. The dosage to the internal sclera for all patients ranged from 404.5 to 601.4 Gy, with a median dosage of 506.5 Gy [Table 1 for more information].

Visual acuity

At the time of diagnosis, of 25 patients with documented visual acuity, 22 patients (88.0%) had a best-corrected visual acuity in the affected eye $\geq 20/70$ (LogMar ≥ -0.54). Twelve months' postirradiation, 14 of 26 patients (53.8%) had maintained a best corrected visual acuity $\geq 20/70$ (LogMar ≥ -0.54), with two of the patients having improved visual acuity. Twenty-four months' postirradiation, 11 of 17 patients (64.7%) had visual acuity $\geq 20/70$ (LogMar ≥ -0.54) with 2 of those patients maintaining 20/20 visual acuity. Two patients had an improved visual acuity without cataract surgery after receiving Ru-106 plaque brachytherapy. No variable was a statistically significant predictor of visual acuity at 1-year follow-up postplaque radiation therapy [Table 3].

Side effects

Three patients, developed cataracts at 20, 37, and <12 months' postradiation, respectively, which after corrective surgery had significant improvement of their best-corrected visual acuity. The one patient who developed radiation retinopathy was able to maintain

a visual acuity of 20/40 at 5 years' follow-up. The one patient who presented with partial retinal detachment before plaque brachytherapy developed total retinal detachment 32 months postradiation. None of the patients had local recurrence of tumor, and there were no enucleations reported at the last follow-up visit. No patient deaths have been reported that were attributable to choroidal melanoma to most recent follow-up.

Discussion

In this retrospective study of a cohort with medium-sized choroidal melanoma; we have shown similar visual acuity and side effect profiles for patients as reported by European radiation oncology centers.^[12,13,15] In North America, several centers have primarily treated medium-sized choroidal melanoma by I-125 plaque brachytherapy,^[16-21] however, Ru-106 is emerging as an effective mode of therapy as well; this study supports consideration of Ru-106 as an alternative radical treatment. In addition to a reasonable postradiation visual acuity and side effect profile,^[14,18] other incentives to use the Ru-106 over I-125 include a thinner plaque (1.0 mm vs. 4.5 mm), easier preparation, and reusability, as described elsewhere.^[19] Furthermore, one study has highlighted that Ru-106 is superior to I-125 plaque brachytherapy in sparing normal intraocular structures.^[18] As a β -emitter, Ru-106 is ideal as it has a smaller depth penetration and thus less damage to surrounding normal tissues.

In our center, we had begun using the Ru-106 plaques in 2001 for tumors equal to or <5.0 mm in apical height and <16.0 mm in maximum basal tumor diameter as the dosimetric characteristics within those dimensions are similar to I-125. Twenty-eight patients have been treated with Ru-106 plaque brachytherapy for choroidal melanoma since the program commenced. Twenty-two patients out of the 25 (53.8%) who had 12 months' follow-up and eleven of 17 (64.7%) patients

Table 1: Baseline patient and tumor characteristics

Characteristic	Value
Gender	
Male	12
Female	16
Age, median (years)	47-93 (67)
Laterality	
Right	12
Left	8
Tumor height, median (mm)	2.3-5.25 (3.5)
≤ 5	25
>5	2
Longest basal dimension, median (mm)	6.3-14.0 (10.65)
Location (of 14 patient records)	
Anterior to equator	4
Posterior to equator	15
Ciliary body melanoma	2
Unknown	7
Treatment time (radiation exposure length), median (h)	57.1-170.9 (98.2)
Dose, median (Gy)	
Prescription point (5 mm)	81.4-89.4 (84.7)
Tumor apex	76.2-268.1 (157.0)
Internal sclera	353.4-660.4 (506.1)
Follow-up, median (months)	1.5-91.5 (31.2)

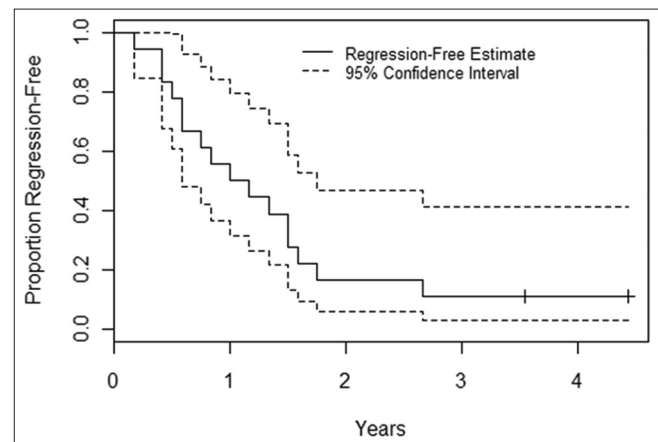


Figure 1: Kaplan-Meier curve showing time until tumor regression. Median time to regression was estimated to be 13 months (7-21)

Table 2: Predictors of time to regression

	Hazard ratio (95% CI)	P
Age	0.98 (0.95-1.02)	0.32
Male	1.04 (0.44-2.46)	0.93
Diabetes	0.39 (0.11-1.32)	0.13
Hypertension	0.50 (0.15-1.71)	0.27
Size (n=20)	1.00 (1.00-1.00)	0.60
Retinal invasion	0.67 (0.15-2.91)	0.59
Coronal (n=20)	0.64 (0.22-1.87)	0.42
Radiation strength	1.10 (0.91-1.32)	0.35
Radiation dose at apex	1.00 (1.00-1.01)	0.33
Treatment time	0.99 (0.98-1.00)	0.14

CI: Confidence interval

Table 3: Predictors of visual acuity at 1 year

	OR (95% CI)	P
Age	0.94 (0.88-1.00)	0.065
Male	1.14 (0.22-5.84)	0.87
Diabetes	0.30 (0.03-3.07)	0.31
Hypertension	0.41 (0.04-4.31)	0.45
Size (n=18)	1.00 (0.99-1.00)	0.15
Coronal	0.17 (0.02-1.42)	0.10
Radiation strength	1.00 (0.70-1.42)	0.98
Radiation dose at apex	1.01 (0.99-1.03)	0.32
Treatment time	1.01 (0.98-1.03)	0.67

OR: Odds ratio, CI: Confidence interval

with 24 months' postirradiation maintained best visual acuity $\geq 20/70$ (LogMar ≥ -0.54). Damato *et al.*, in a series of 458 patients, reviewed the effect of visual acuity after Ru-106 brachytherapy with median length and thickness dimensions of 10.6 mm and 3.2 mm, respectively, found an actuarial probability of 20/40 or better was 0.80 and 0.93 for 20/200 or better at 24 months' follow-up. In another series of 100 patients with choroidal melanomas not exceeding 5.0 mm and 15.0 mm in height and base respectively, the authors found a success of maintaining a visual acuity equal to or better than 20/70 and 20/200, 36 months' postirradiation of 27% and 41%, respectively.^[22] In addition, in our study, two patients had an improved visual acuity, and two more patients had improved vision compared to presentation after postradiation cataract removal.

A concern with radiation treatment is local recurrence and/or progression of tumor, which in our series of patients has not occurred to date of most recent follow-up. Of 25 patients with documented posttreatment dimensions, 25 had a regression in tumor dimensions. The side effect profile of our series of patients at 24 months' postirradiation included three patients experiencing a decrease in visual acuity; three patients developed a cataract in the treated eye that with surgery resumed stable visual acuity in two and pending follow-up in the third. There were no reported cases of secondary plaque brachytherapy, enucleation, or death secondary to the melanoma for our patient population. This is comparable

to a study on irradiation-related complications after ruthenium treatment for choroidal melanomas by others.^[23] in which they reported after 36 months' follow-up of 100 patients an incidence of radiation-induced cataracts of 28%, nonneovascular glaucoma 91%, and enucleation and radiation-induced optic neuropathy both in 10% of their patient population. Furthermore, prior studies^[24,25] have suggested that there is a direct relation between tumor height and visual loss. In our study, we did not find this to be a significant relation.

For choroidal melanomas, calculating dose to the sclera accurately can be challenging. The treatment planning system used at our center was to validate the prescription point dose. All other data were provided by observation only. We were able to capture 27 of 28 patients' sclera doses according to the treatment planning system algorithm. One patient's sclera dose was not able to be obtained as there was intraoperative difficulty involving an extraocular muscle and plaque placement. A second patient had their sclera dose calculated through a nonstandard algorithm, which was then revised relative to the others.

While acknowledging that our study has a small number of patients, our results supports those findings from the larger European studies who have reported on Ru-106 plaque brachytherapy for choroidal melanoma. From a North American perspective that is largely familiar with I-125 plaque brachytherapy, our experience suggests Ru-106 plaque brachytherapy as a safe and efficacious alternative for choroidal melanoma treatment of tumors equal to or <5 mm in apical height. Future considerations are a longer follow-up and collection of a larger database of patients to further determine the efficacy and safety for Ru-106 plaque brachytherapy for medium-sized choroidal melanoma treatments.

Conclusion

In this study, we found that Ru-106 plaque brachytherapy is an efficacious and safe treatment option for patients with medium-sized choroidal melanomas.

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Conflicts of interest

There are no conflicts of interest.

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